

Knowledge Graph Augmented Multilingual Benchmark for Factual LLM Evaluation

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Abstract. Large Language Models (LLMs) demonstrate high fluency on natural language generation while still being vulnerable to hallucinations and factual inaccuracies. Current evaluation benchmarks are English-centric and not grounded in structured knowledge, which keeps them from fully capturing the factual reliability of LLMs across languages. In this research, a Knowledge Graph enhanced multilingual benchmark is proposed to assess the factual consistency of LLM outputs. The benchmark combines subgraphs of domain-specific knowledge graphs and varied prompt templates to facilitate structured, context-sensitive testing for multiple Indic and world languages. Evaluation criteria are also suggested for assessing accuracy and cross-language robustness. The aforementioned model is designed to offer a huge data source for validating and enhancing the factual reasoning abilities of LLMs which may be used for multilingual information retrieval, dialogue systems, and other knowledge-based applications.

Keywords: knowledge graph, large language models, multilingual evaluation, factual accuracy, natural language processing

1 Introduction

The success in natural language processing tasks by Large Language Models (LLMs) has been remarkable in recent years. Nevertheless, even with the proficiency exhibited by these models, hallucinations or erroneous output is common for LLMs in certain scenarios, particularly in applications requiring a lot of knowledge. Concerns remain regarding the accuracy and reliability of LLMs in practical applications like healthcare, law, and multi-lingual information searches.

The existing approaches to factuality evaluation are generally English-oriented and highly dependent on the unstructured comparison of texts. Traditional factuality evaluation measures such as BLEU and ROUGE have not been very successful in measuring semantic correctness and factuality in multilingual and

low-resource language settings like Indic languages. Furthermore, there is little presence of structure in the data used in existing benchmarks, which reduces their effectiveness in factually validating text.

This study proposes an approach to multilingual factuality evaluation based on knowledge graphs. The integration of structured subject-predicate-object (SPO) triples, multilingual prompt templates, and domain-specific knowledge graphs enables the development of a scalable and interpretable evaluation framework. This factuality benchmark, as opposed to the existing ones, will focus on factuality validation and hallucination identification in multiple languages.

The proposed study aims to address the following research questions:

- **RQ1:** In what ways can KGs help with the improvement of factuality verification in LLMs for different languages?
- **RQ2:** How well do evaluation methods based on KGs perform against classical metrics of lexical overlap like BLEU and ROUGE?
- **RQ3:** What level of factual consistency can LLMs maintain in multilingual and low-resource language settings?

The major contributions of this work are summarized as follows:

- A multilingual Knowledge Graph benchmark with diverse multilingual prompts for factual evaluation is proposed.
- Support is provided for Indic languages and other world languages to enhance cross-linguistic factuality evaluations.
- An evaluation metric to measure factual accuracy and hallucination resilience is proposed.
- A scalable system to generate multilingual benchmarks from canonical SPO triples is developed.

2 Literature Review

One area that has gained prominence is the integration of Large Language Models (LLMs) and Knowledge Graphs, mainly focused on overcoming some key gaps that exist within LLMs, like inconsistencies (Hallucinations), uninterpretable reasoning, and reasoning about domain knowledge [17]. The papers surveyed through this collection look at a variety of approaches and architectures to improve LLMs’ capabilities on question-answering and dialogue generation, by exploiting the well-structured and verifiable knowledge offered by Knowledge Graphs.

2.1 Mitigating Hallucinations and Enhancing Factuality

Integration of LLMs and KGs to overcome problems of hallucinations, uninterpretable reasoning, and domain-specific inconsistencies has attracted much attention recently [17]. Previous studies have considered several architectures

that can help improve Q&A and dialogue generation by leveraging structured and verifiable knowledge from KGs.

The authors of FSPO [14] designed a step-wise policy optimization algorithm that incorporates fact-checking at intermediate reasoning steps to alleviate hallucinations. The work emphasizes the significance of grounding not only final answers but also the entire reasoning process in verifiable facts, especially in light of the unreliability of generated explanations in textual reasoning tasks [26].

Another example is ChatTf [22], which leverages domain-specific KGs and RAG [13] to mitigate hallucinations in folklore-based dialogue systems. CDT [24], on the other hand, uses truthful and hallucinatory comparator models to guide token generation during decoding. Another method, BEINFO [19], utilizes behavioral fine-tuning to enhance factual consistency in dialogue systems. Wang et al. [20] proposed an advanced KG-enhanced fact verification system based on graph representations of structured data.

2.2 Dynamic Methods

Two papers addressed dynamic methods of real-time knowledge integration during the inference stage, which aligns with broader knowledge-augmented NLP methodologies [12]. In particular, Baek et al. [5] present the Knowledge-Augmented framework, a zero-shot approach to prepending salient facts to the LLM’s input prompt after retrieving them from a Knowledge Graph. This method supports the model’s output by referring to information from outside the context, which has been shown to greatly boost the performance in Knowledge Graph Question Answering (KGQA) tasks and multilingual local knowledge benchmarks [11].

The Dynamic source Grounded decoding (DoGe) method [23] properly balances between extracting features and balance between the often conflicting goals of presenting varied and interesting answers and maintaining factuality through dynamic switching, depending on an LLM’s knowledge and external Knowledge Graph data as a function of the model’s factual confidence.

Zheng et al. [28] present a Knowledge Graph guided Retrieval-Augmented Generation framework that integrates structured graph retrieval with LLM generation. Their work enhances contextual relevance, consistency in facts, and multi-hop reasoning.

Opsahl [16] demonstrates that a fine-tuned BERT can actually outperform models which are more complex and dynamically trained. Similar diagnostic frameworks have also been explored for hallucination analysis in Arabic machine reading comprehension systems [2].

2.3 Grounding Reasoning and Complex Question Answering

Amayuelas et al. [3] propose that the data derived from the knowledge graph (KG) can be associated with every step involved in the LLM’s process reasoning chain, such as Chain-of-Thought (CoT) and Tree-of-Thought approaches. Their work indicates a marked improvement in terms of question answering, highlighting the benefits of clear, verifiable, and sequent reasoning [25].

Qi et al. [18] present the framework of IntelliChain for Socratic dialogues in educational environments. To ensure both accuracy and educational efficacy, especially with complex problem-solving in mathematics and related fields, their system integrates a chain-of-thought approach in multiagent architecture through a knowledge graph.

2.4 Knowledge Graph Format Driven Approaches

Semantic and structural information is incorporated in knowledge graphs to build the reasoning paths by considering the most relevant reasoning paths first [21]. The approach offers effective guidance for LLMs to perform complex multi-hop inference.

An empirical study by Dai et al. [9] of LLM preferences shows that large scale models work better with unordered, linearized triples rather than fluent natural language text. This result can be effectively utilized where a biomedical question-and-answer system uses LLMs for translating natural language questions into a computationally processable form [1].

In [8], Chen et al. exploit the power of Knowledge Graph-augmented large language models to generate answers to biomedical questions. Their method leverages domain knowledge to provide accurate answers and minimize hallucinations.

Chen et al. [7] introduce Knowledge Graph-Driven Supervised Fine-Tuning (KG-SFT), which improves a model’s capacity for knowledge manipulation and reasoning. In low-data, knowledge-intensive domains, KG-SFT significantly improves performance by augmenting the training data with high-quality, logically coherent explanations from Knowledge Graphs.

2.5 Architectural Innovations and Advanced Methodologies

There exist several novel frameworks that attempt to integrate LLMs and knowledge graphs. Erickson et al. [10] present the ChatBS-NexGen architecture with domain-specific Knowledge Graphs for improving management, repeatability, and verifiability.

Zhang [27] explains a complete pipeline regarding the augmentation of LLMs using accurate Knowledge Graphs. Ao et al. [4] propose LightPROF, a “light-weight framework” that utilizes a knowledge adapter to encode both semantic meaning and world knowledge into textual and structural information extracted from a Knowledge Graph into a “knowledge soft prompt”.

The study conducted by Liu et al. [15] centers around the enhancement of Large Language Model trustworthiness by utilizing Knowledge Graphs. The study provides evidence that the use of structured external knowledge contributes to better factual grounding and interpretation.

Bai et al. [6] demonstrate using LLMs to build the Knowledge Graph for framework material collections, which leads to a considerable increase in accuracy of a question-answer system in this field.

3 Methodology

The Knowledge Graph Augmented Multilingual LLM Evaluation Benchmark is developed using a comprehensive multi-step methodology. The process converts messy raw multilingual text into a formatted, verified Knowledge Graph that can be utilized to measure an LLM’s performance. This methodology emphasizes quantitative measurements such as Factual Accuracy, Resistance to Hallucination, and Intrinsic Cross-Language Consistency, as opposed to relying on the traditional linguistic measures of surface-level.

3.1 Benchmark Design Principles

These principles are formed around the primary limitation that traditional methods of Quantifying Factual Integrity Through Lexical Overlap (such as the standard F1 score) have not provided an accurate way of evaluating factual accuracy when applied to multilingual paraphrases and fluent translations. The proposed benchmark establishes a Knowledge Graph-grounded framework for objective factual evaluation.

- **Canonical Factual Representation:** Knowledge will be represented as Language-Independent Subject–Predicate–Object (SPO) triples. Therefore, it will provide a Canonical Ground Truth not influenced by the Query Language (i.e. Hindi, English, etc.) used to query the LLM.
- **Decoupling Content and Language:** The architecture intentionally separates the core factual structure (stored as structured graph representations of entities and relations) from its textual realization (stored as multilingual surface-level factual statements). This design allows the system to validate the semantic correctness of an LLM’s output by matching extracted entities (regardless of language) back to the canonical identifiers defined within the knowledge graph.
- **Factuality Over Fluency:** Evaluation prioritizes measures of correctness and non-contradiction (Hallucination Rate, Factual Faithfulness) over simple linguistic similarity, a critical requirement for high-stakes domains like Legal and Biomedical analysis.

3.2 Data Collection and Knowledge Construction

The corpus is built using multilingual, domain-specific sources that require normalising them all and transforming them into a unified graph structure. The knowledge developed adheres to a strict seven-step cyclical process:

1. **Raw Data Gathering:** The raw data available in multiple languages, such as Law, Drug–Illness Relations, and Sports Facts, is gathered from structured domain-specific repositories that are designed for factual evaluation in specific domains.

2. **Data Cleaning and Preparation:** The raw text data is standardized through a preprocessing module that includes linguistic segmentation, noise removal, and dialect standardization. This step is crucial for ensuring that the raw data gathered is of the required quality for structured information extraction.
3. **Semantic Triple Extraction:** The initial transformation of the raw text data is carried out through a specialized triple extraction and relation mapping module. The raw text data is analyzed to identify entities and relations, which are then used to form canonical subject–predicate–object (SPO) triples that are suitable for structured knowledge graph representation.
4. **Knowledge Consolidation and Benchmark Generation:** The generated triples are merged with the existing knowledge base and validated by converting them into an improved knowledge graph format. This merged representation is maintained in both machine-readable and tabular formats, along with a canonical triple repository, which is used for benchmarking purposes.
5. **Multilingual Query Construction:** This is an important intermediate step where the inputs for the language models are generated.
6. **LLM Evaluation Round:** The queries are evaluated using an evaluation pipeline developed on top of the chosen large language models. This step uses appropriate model checkpoints and configuration parameters to ensure controlled, consistent, and repeatable benchmarking under different experimental settings.
7. **Performance Metrics Calculation and Analysis:** The results produced by the LLMs are evaluated against the gold-standard knowledge graph triples to compute performance metrics. The evaluation framework performs a thorough comparison, alignment, and scoring analysis, and the results are systematically tracked to ensure reproducibility and benchmarking reliability.

3.3 Multilingual Query Generation

To properly assess the multilingual factual correctness of the LLM, the test task set should be able to express the same semantic meaning regardless of the language of input.

Conversion from Triple to Natural Language The LLM model has a dedicated transformation module that translates the standardized Subject-Predicate-Object (SPO) triples, each identified by unique canonical entity identifiers such as QIDs or URIs in the unified Knowledge Graph, into natural language question templates or fill-in-the-blank queries in different target languages.

Ensuring Semantic Equivalence The suggested approach ensures that a question about a specific subject (such as the capital of a country) asked in English and its semantically equivalent translation in a different language (French,

for example) points to the same specific piece of factual knowledge in the Knowledge Graph. This is an important requirement for evaluating multilingual factual consistency and retrieval efficiency.

3.4 LLM Evaluation Protocol

The framework of assessment has been designed with a focus on scalability and automation for purposes of benchmarking, where the quality of the response is tightly linked to the facts represented in the knowledge graph.

- The Multilingual Prompt Generator is responsible for providing the LLM inference module with a vast set of semantically similar but lexically distinct questions in multiple languages, thus enabling a comprehensive evaluation of cross-lingual consistency and factuality.
- The generation procedures are carried out in strictly controlled inference environments, using standardized configuration parameters to ensure consistency across different experiments. All model outputs are carefully documented and archived for subsequent analysis and evaluation.
- The evaluation phase further leverages state-of-the-art entity resolution and linking techniques to match the LLM’s automatically produced, possibly paragraphed, or culturally adapted entity responses with their canonical forms in the reference Knowledge Graph, thus enabling a comprehensive assessment of factuality as opposed to text similarity.

3.5 Evaluation Metrics

The performance evaluation is carried out through a dual-metric system, which integrates traditional metrics for a linguistic baseline with more sophisticated scores based on Knowledge Graph-Centric metrics, which are more focused on fact-checking.

Traditional Linguistic Metrics (Baseline)

- **Exact Match (EM):** A binary evaluation metric that assesses token-level exactness between the predicted answer and the reference answer. The score is 1 if the predicted response matches the gold-standard answer after undergoing normalization (such as lowercasing and removing punctuation), and 0 otherwise. This metric assesses exact factual accuracy but ignores semantically similar alternatives.
- **F1 Score:** This is a token-level metric calculated by computing the harmonic mean of precision and recall. Precision is the ratio of generated tokens that are relevant (i.e., included in the reference answer), while recall is the proportion of reference tokens that are actually captured in the predicted response. The F1 score gives partial credit for overlap and is especially helpful when the model predicts answers that are paraphrased but somewhat correct.

- **Factual Precision (FP):** An evaluation metric that is knowledge-based and aims to measure factual precision rather than focusing on text similarity. This metric evaluates not only token-level overlap but also whether the output entities and relationships match the subject-predicate-object triples in the reference knowledge graph. This metric is particularly important when it comes to knowledge-based question-answering tasks, where semantic precision is more important than text similarity.

3.6 Benefits of the Proposed System

The achievement of a unified knowledge graph integration is one of the most important methodological breakthroughs in the evaluation of multilingual LLMs, and it leads to three major scientific and engineering achievements:

- **Enhanced Objectivity and Traceability:** By focusing exclusively on strictly validated Subject-Predicate-Object triples as the sole source of truth, the evaluation acquires an objective, fact-based perspective on model performance. Each model response can be traced to a canonical representation of factual knowledge, which significantly improves the traceability of model behavior.
- **Cross-Lingual Fairness:** The Multilingual Evaluation Schema provides an efficient framework to distinguish the impact of language-specific lexical assessment metrics in the evaluation framework. Models are judged exclusively on their ability to access and state the same set of facts in different languages, thereby evaluating models based on factual knowledge gaps while remaining tolerant of correct linguistic and cultural paraphrasing.
- **Detailed, multi-faceted assessment:** The Overall Grounded Performance Index is a set of three independent, measurable, and easily interpretable axes (factual groundedness, robustness to hallucination, and cross-lingual robustness), which is vital for developing trustworthy systems deployable worldwide in high-stakes, knowledge-intensive domains.

4 Result Analysis

This section provides a detailed analysis of the outcome regarding the instruction tuning of this language model, which is enhanced by a multilingual knowledge graph. The analysis focuses on both the quality of the generated output and factual consistency, including the primary tasks of providing multilingual support and fact-based knowledge.

4.1 Overall Performance Summary

Table 1 presents the primary evaluation metrics on the validation and test sets.

These results demonstrate good training convergence with a small difference between the training and validation losses. This indicates good generalization performance and a lack of overfitting, which is especially true given the complex nature of the multi-lingual and multi-task dataset.

Table 1. Overall Model Performance Across Key Metrics

Metric	Score
Training Loss	1.20
Validation Loss	1.30
BLEU	56.0
ROUGE-L	0.85
Exact Match (EM)	0.45
Triple F1	0.65

4.2 Loss Analysis

The fact that the trained loss is achieved at no more than 1.20, while the validation loss is no more than 1.30, emphasizes the efficient optimization process. Given the innate complexities involved within the task, including multilingual text processing, knowledge alignment according to structured knowledge, as well as the proficiency of processing knowledge from both the KG-to-Text and Text-to-KG aspects, the trained loss is evidence that the trained model is capable of understanding the knowledge graph triples and their representation within natural language form.

The small gap between training and validation losses again emphasizes strong generalization capabilities over new and unseen data as well as for different languages.

4.3 Generative Text Quality Evaluation

BLEU Score. A BLEU score of 56.0 indicates strong performance in multilingual knowledge-grounded generation tasks like paraphrasing and factual text generation. The high overlap between the n-grams and the reference texts is a sign of the model’s ability to retain the essential lexical parts while generating fluent and diverse texts. The BLEU results further indicate effective processing of predicate semantics and entity relationships between entities obtained from the input triples in knowledge-to-text tasks.

ROUGE-L Score. The model attained a score of 0.85 in the ROUGE-L evaluation metric, which signifies perfect similarity in the longest common subsequence with regard to the reference text. This is significant for preserving the logical structure of the generated output with regard to entities, relationships, attributes, and other factual aspects of the narrative.

4.4 Factuality and Structured Prediction Analysis

Exact Match (EM). A score of 0.45 in the exact match indicates the proportion of times the predicted and actual outputs match exactly, which is particularly important for the Text-to-KG task. The fact that there are challenges involved

in exact matching, coupled with the considerations of multilingualism, paraphrasing, and possibly format variations, achieving an exact match score of 45% demonstrates strong structured prediction capability. These results indicate that a significant part of the structured outputs matches exactly with the reference, which is a clear indication of the high level of accuracy involved in the extraction and recreation process of triples.

Triple F1 Score. The Triple F1 measure of 0.65 shows good entity and relationship extraction from generated triples. This is because the system ensures that the semantics of relationships are maintained even without matching.

4.5 Why BLEU/ROUGE May Fail While KG-Based Metrics Succeed

While BLEU and ROUGE-L scores provides useful insights into the lexical and structural level of similarity, they are not always reliable measures of factual correctness. On the other hand, metrics that are aware of the KG, such as Exact Match (EM) and Triple F1, are designed to measure the alignment of structured facts.

Limitations of BLEU and ROUGE in KG-Grounded Tasks. BLEU and ROUGE are surface level overlap scores. They are designed to calculate n-gram precision (BLEU) and subsequence recall (ROUGE-L) between a generated text and a reference text. Although they can be useful in summarization and translation tasks, they have several flaws when applied to a structured factual domain:

1. **Sensitivity to Paraphrasing:** BLEU penalizes semantically equivalent paraphrases that use different lexical forms. Despite being factually identical, n-gram differences reduce BLEU score.
2. **Surface-Level Evaluation:** These metrics do not explicitly verify relational correctness. A sentence may achieve high BLEU overlap while containing a critical factual error.
3. **Multilingual Variability:** In a multilingual environment, the lexical realization varies greatly across languages. The morphological complexity, word order, and transliteration variations further decrease the overlap of n-grams, even when the semantic meaning is maintained.
4. **Formatting Sensitivity in Structured Tasks:** In Text2KG tasks, slight variations in formatting can cause a drastic reduction in BLEU/ROUGE scores even when triples are correctly extracted.

Why Knowledge-Graph Metrics Perform Better. Knowledge-graph-oriented metrics, such as Exact Match and Triple F1, directly evaluate structural correctness rather than surface-level text similarity. The primary distinction lies in the evaluation objective of each metric:

- BLEU/ROUGE: Measure lexical overlap.

- KG Metrics: Measure relational correctness.

In knowledge graph based tasks, the task is considered correct if the relational fact is correct, irrespective of whether the words in the sentence match the reference sentence. As a result, the KG-based metrics are better aligned with the genuine task of ensuring factual accuracy.

4.6 Interpretation and Discussion

Cumulatively, the above findings not only demonstrate the effectiveness of the multilingual instruction tuning model in achieving a balance between textual fluency and factual accuracy but also its ability to process structured knowledge, as evident from its performance on Exact Match and Triple F1 scores.

These findings are particularly significant in the multilingual scenario and with the evaluation metric being based on unseen relations, which indicate that the model has the ability to comprehend generalizable relational concepts rather than just recognizing surface-level patterns, which is essential for scalable multilingual factual reasoning.

4.7 Limitations and Error Patterns

In spite of the impressive results, analyzing the performance of the Triple F1 against the Exact Match metric reveals some partial errors. These errors appear as a function of inconsistencies regarding predicate normalization, cross-lingual entity variations, and the differences in the formatting of structured outputs.

5 Conclusion

The benchmark has been created using a Knowledge Graph to test the efficacy of LLM grounding in providing accurate factual knowledge. The existing benchmarks rely mostly on the English language, consisting of mostly unstructured data, which is not suitable for multilingual settings or identifying tiny hallucinations. Structured Subject-Predicate-Object triples provide an effective framework for testing facts in a multilingual setting, using an independent model that verifies the validity of information and is called the ‘Ground Truth’.

The current benchmark consists of multilingual pre-processing, triple canonicalization, automatic context creation when generating the answer based on the information stored in the Knowledge Graph, as well as creating training data for KG-to-text and text-to-KG models. The proposed framework enables factual assessment in Indic and global languages.

5.1 Limitations

However, it is necessary to take note of certain limitations associated with the proposed approach prior to any further generalization or utilization of the results presented above.

Theoretical Limitations From a theoretical perspective, the proposed benchmark has its inherent limitations since it is based on the use of a static representation of knowledge using subject-predicate-object triples. In other words, the benchmark does not incorporate the possibility of accounting for the temporality, contextuality, or probability of real-world facts. In this regard, a triple defining some medical or legal relationship can lose its validity at some point in the future; however, the benchmark cannot account for this fact.

Additionally, it should be noted that the metrics used to assess the quality of reasoning are focused on structural matches rather than the semantics of natural language answers. Consequently, an agent can provide a semantically valid response containing a factually correct answer but receive a low score due to a difference in the surface representations of entities and predicates used in the question.

Also missing from the benchmark is multi-hop reasoning through the Knowledge Graph. Factual questions that necessitate traversing multiple relational edges, such as finding out the agency responsible for regulating a drug that is effective in treating a disease that affects a specific demographic, cannot currently be tested by the evaluation method under consideration.

Practical Limitations Moving towards practical issues, the biggest limitation of the current implementation is that the language coverage is limited to low-resource and Indic languages. Low-resource languages like Marathi, Bengali, Hindi, and Arabic suffer from near-zero accuracy since the current model suffers from tokenization problems related to non-Latin alphabets and a lack of fine-tuning data. This becomes a problem since these are hundreds of millions of people who depend on factual correctness in AI-powered systems.

In terms of the current infrastructure, which has been developed and evaluated using Jupyter Notebook and Google Colab, the biggest limitation is that there is no publicly available dataset and no RESTful API and leaderboard to evaluate the system.

Furthermore, since the knowledge graphs employed in this research have domain restrictions such as Biomedical, Legal, and Sports domain KGs, there is an element of selection bias as well as restricted generalizability beyond those high-stakes domains to others such as Finance, Climate policy, or Public Administration. It is important to note that the quality of evaluation in this task is contingent upon the correctness of the underlying KG. Gaps or errors in the KG result in erroneous ground truth assessments.

The third limitation is related to the benchmark not evaluating reasoning faithfulness or whether the response generated by the model has any connection to the retrieved triples. In other words, whether the response generated is based on parametric memory only and not the knowledge graph at all.

5.2 Future Work

These limitations also highlight several opportunities for future research. Firstly, expanding the scope of language support to over 50 languages, with special

emphasis being given to Indic scripts. Creating custom instruction datasets and use of Indic tokenizers like IndicBERT and MuRIL

In order to compensate for the structure-related limitations of the SPO format, the future versions of the benchmark will experiment with the integration of Graph Neural Network (GNN)-based encoders like Relational Graph Convolutional Network (R-GCN) to produce topology-sensitive embeddings for injecting into the LLM cross-attention layers. Other KG embedding models like TransE and RotatE will provide the means for performing entity matching across KGs in a common semantic space regardless of string overlaps.

The issue with the static triple retrieval process may be addressed through designing a dynamic evaluation pipeline integrated with live endpoints like Wikidata SPARQL API together with dense retrievers (DPR, Contriever). The benchmark will thereby allow for evaluating newly emerged claims while being able to analyze the border between parametric and retrieved knowledge.

From the perspective of the technical infrastructure, the benchmark’s dataset will be hosted on the Hugging Face Hub as an openly available resource, together with RESTful evaluation API and a leaderboard for monitoring the cross-lingual factuality scores.

The domain coverage will be expanded from the existing three domains, adding Finance, Climate Science, and Public Policy. Furthermore, multi-hop reasoning tasks will be required that involve traveling through two or three relational edges and incorporating distractor subgraphs to assess robustness against adversarial facts. Finally, reasoning faithfulness will be tested via a modified RAGAS framework, with the created dataset being used for training lightweight LLMs using a KG-based consistency reinforcement learning reward scheme, explicitly reducing hallucinations.

As LLMs become increasingly widely deployed on a global scale in various domains and languages, a benchmarking framework of this type is no longer just beneficial but becomes necessary to ensure factual accuracy and robustness.

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